Kyle Ton

San Diego, CA | tonkyle06@gmail.com | www.kylehton.com | linkedin.com/in/kylehton | github.com/kylehton

Education

University of California, San Diego Expected Jun. 2027
B.S. in Computer Science and Mathematics *La Jolla, CA*
- **Relevant Coursework:** Data Structures and Algorithms, Computer Architecture and Assembly, Linear Algebra and Differential Equations, Numerical Optimization of Linear Systems, Graph Theory

Experience

Software Development Engineer Intern Jun. 2026 – Sep. 2026
Amazon Web Services - AWS Security Assurance *Seattle, WA*
- Saved an estimated **225 user-hours annually** across **450 security issues/year** by building and shipping an internal triage agent with Strands, Amazon Bedrock, and Bedrock Knowledge Bases that reduces triage time by **0.5 hours per issue**
- Achieved **70.4% root-cause classification accuracy**, **100% enqueue success**, and **zero observed failures** in beta/gamma by architecting an idempotent event-driven pipeline with DynamoDB Streams, EventBridge Pipes, SQS, Lambda, and API Gateway
- Reduced exposure to prompt attacks and over-privileged access by hardening AI generation with **Bedrock Guardrails**, nonce fencing, character sanitization, HTML/link flattening, and granular least-privilege IAM across agent operations and resources

Software Engineering Intern - Full-Stack Jun. 2025 – Aug. 2025
ForOurLastNames *Remote, US*
- Led end-to-end production deployment of a **containerized Node.js application** by designing AWS infrastructure (Lambda, S3, CloudFront, API Gateway), enabling stateless horizontal scaling and **sub-200ms** API latency under internal load testing
- Implemented a **production-grade authentication and billing pipeline** using Supabase Auth and Stripe, enforcing Row-Level Security policies and verified webhook handling to manage user-scoped access across 50 test accounts and ensure idempotent subscription states under repeated webhook deliveries
- Designed and deployed **observability dashboards** using Prometheus, Grafana, and CloudWatch metrics to track Lambda invocations, cold starts, and memory utilization, enabling data-driven capacity planning and infrastructure tuning

Software Engineering Intern - Full-Stack + AI Oct. 2024 – Dec. 2024
PM Accelerator *Remote, US*
- Contributed to backend development of a **FastAPI-based internal tool** integrating an NLP model, evaluating resume keyword coverage across 100 test documents and validating a **15%** increase in matching keywords
- Architected an **asynchronous task processing system** using Celery and Redis, **parallelizing LLM workloads** and reducing end-to-end processing latency by **50%** (20s to 10s) during internal evaluation
- Designed **MongoDB schemas** for resume, job description, and evaluation result collections, structuring nested keyword-match metadata and indexing on user and document IDs to support fast retrieval across concurrent evaluation runs

Projects

Labl | *Python, FastAPI, MongoDB, Gmail API, Celery, Redis, AWS EC2, HuggingFace Sentence-Transformers*
- Engineered a hybrid email classification pipeline combining **BAAI/bge-small-en-v1.5** dense embeddings with **BM25 sparse retrieval**, driving an adaptive classifier that bootstraps from a single medoid and **matures to k-means clustering** upon reaching a confirmed label threshold
- Built a **distributed Celery Beat + Redis automation layer** scheduling per-user inbox scans, authenticating worker-initiated runs via decrypted OAuth refresh tokens with a pre-enqueue watermark preventing duplicate execution
- Implemented a **real-time feedback loop** propagating user label confirmations as ground truth to synchronously **refit per-user cluster medoids** for each specified label, improving accuracy across sessions

Git-Lint | *Python, FastAPI, Pinecone, AWS Lambda/S3, Docker, OpenAI Agents SDK*
- Built an **event-driven code review agent** triggered by **GitHub pull-request webhooks**, analyzing **incremental diffs** to minimize processing cost and reduce end-to-end latency
- Designed a **repository-scoped embedding retrieval pipeline** using **Pinecone and S3** to surface contextually relevant code across **200+ indexed files** without loading the full repository into memory
- Implemented **idempotent webhook ingestion** with **deduplication** and **retry logic**, on a Docker and AWS Lambda deployment, resolving cold-start and timeout constraints to maintain predictable response behavior under invocation

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, HTML/CSS (Tailwind)
Frontend & Backend: React.js, Next.js, Spring Boot, FastAPI, Node.js, Express, MongoDB, PostgreSQL, REST API, Figma
Cloud & DevOps: AWS (DynamoDB, SQS, API Gateway, Lambda, S3, CloudFront), Docker, GitHub Actions (CI/CD), Redis
Machine Learning/AI: Scikit-learn, XGBoost, RAG Pipelines, Vector Databases (Pinecone)